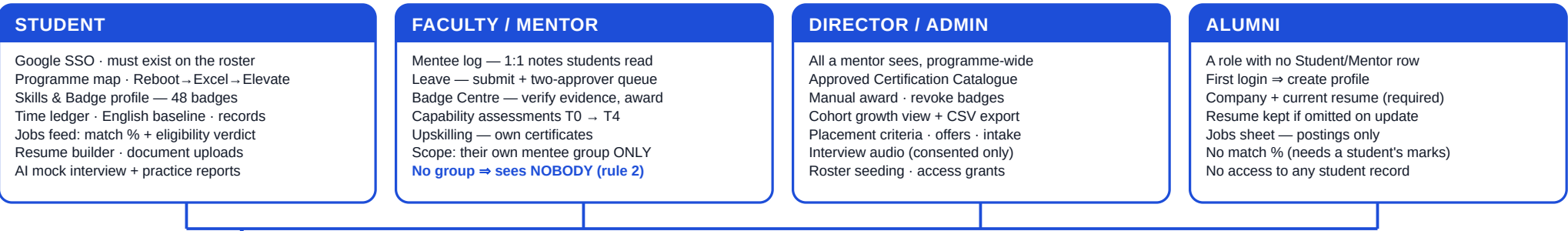


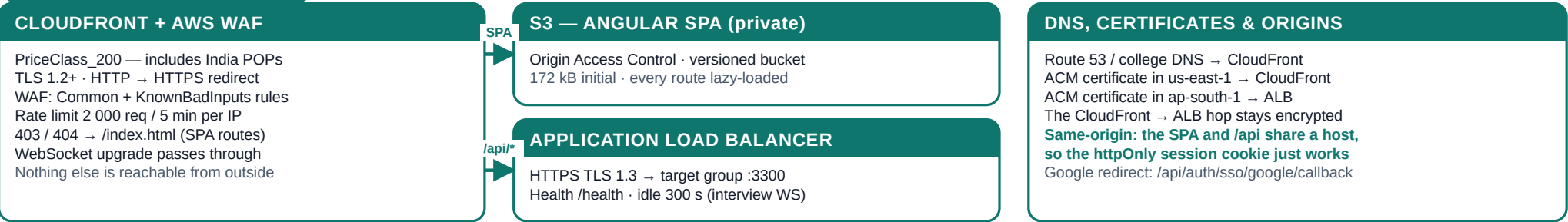
REEP — Placement-Readiness Platform

Complete system architecture, as deployed on AWS

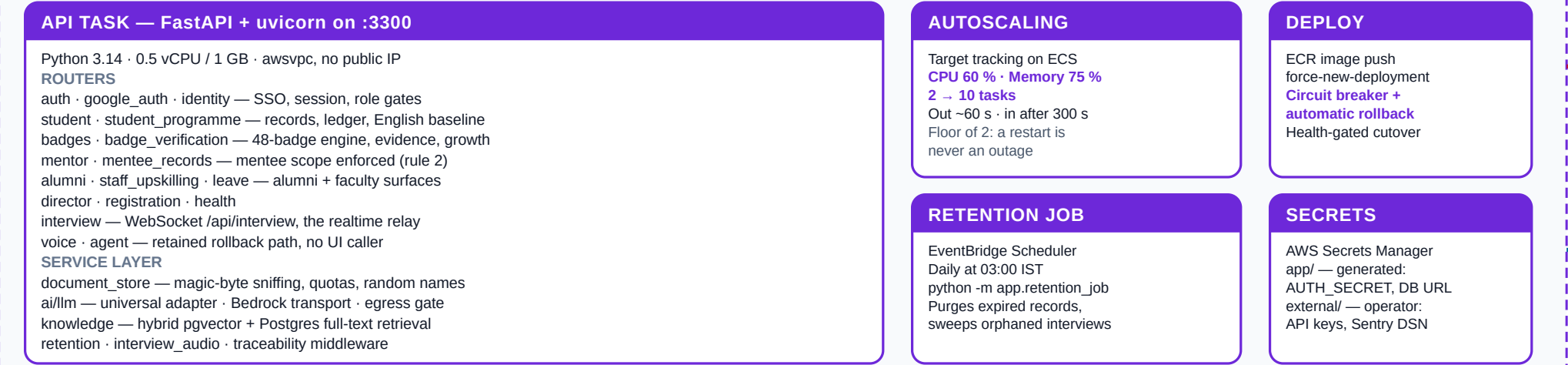
1 · PEOPLE & ROLES



2 · EDGE — ONE PUBLIC DOOR



3 · COMPUTE — ECS FARGATE CLUSTER (private subnets, 2 AZs)



4 · STATE



AUTOSCALING, CAPACITY & COST

API 2 → 10 tasks · RDS storage 20 → 100 GB
RDS compute is FIXED — resizing is a human call
CloudFront · S3 · EFS · ALB scale themselves
≈ **\$130 / month infra before model tokens**
A voice interview costs ≈ \$3–5 — the real variable

BACKUPS & DISASTER RECOVERY

RDS automated snapshots — 14-day point-in-time
AWS Backup vault — daily, kept 35 days
...covering BOTH the database and EFS
Deletion protection + final snapshot on the instance
Two independent planes: one destroy loses nothing

SECURITY & ACCESS CONTROL

Google-only SSO — the roster IS the allowlist
google_sub pins identity · token_version revokes
RULE 1 student PII reaches no model unless opened
RULE 2 staff scope by role, never by a missing field
Boot guard refuses to start on repo-default secrets

5 · AI & MODEL PLANE — called by the API tasks

AMAZON BEDROCK — NOVA

Model: apac.amazon.nova-pro-v1:0 (configurable)
Transport: converse / converse_stream via boto3
Powers: resume brief · grounded assistant · scoring
Auth is the IAM task role — no API key anywhere
In-account: your traffic is not used for training
Rule 1 still applies — student PII needs the flag

OPENAI REALTIME — THE AI MOCK INTERVIEWER

WebSocket /api/interview · 24 kHz PCM both ways
The relay owns the turn: create_response = false
Word-count gate, then exactly ONE response.create
Phases: opening → probing → deep_dive → wrap_up
4 specialisations (HR · DM · BA · FA), voice per track
Verdict spoken, then a strict-JSON scorecard (text only)
Persists sessions · turns · evaluations, fire-and-forget
Consent gate: no live grant ⇒ 4013 · revoked ⇒ 4014
MIGRATION → Amazon Nova Sonic (bidirectional stream).
IAM already allows it; the engine port is still to do.

KNOWLEDGE BASE & RETAINED PATHS

KB: pgvector cosine blended with full-text, distance-floored
Retained (no UI caller): LiveKit voice worker + orchestrator

6 · OBSERVABILITY & TRACEABILITY — fed by every request

SENTRY — THE ONE TOOL

API (Python SDK) + SPA (lazy chunk) → one org
Errors AND performance traces in one place
PII OFF — no cookies, no bodies, no student text
Every event tagged request_id = X-Request-ID
Claude reads it through the Sentry MCP connector

CLOUDWATCH — WHAT SENTRY CANNOT BE

/reep/api log group (30 d) — one access line per request
Alarms → SNS email: no healthy task · ALB 5xx · RDS
storage & CPU · CPU still pegged at max tasks
AI tripwire: “Dropped interview turn” metric filter

THE TRACE THREAD — ONE ID, FOUR HOPS

1 middleware mints or keeps X-Request-ID · 2 echoed on the response · 3 tagged on the Sentry event · 4 printed as rid=...
in the access log — ALB logs in S3 close the loop

7 · INTERVIEW CALL RECORDING

TWO TRACKS, NEVER MIXED

One WAV for the student, one for the AI — kept separate
because the directions are not time-aligned · size-capped

CONSENT IS A ROW, NOT A SETTING

Two switches: the deployment flag AND the student's own
store-audio tick · DIRECTOR/ADMIN only · deleted after 180 days